

THE FACT THAT DECIDES THE ARCHITECTURE

Arabic is a supported input language and not a supported output voice.

AssemblyAI's managed Voice Agent API, from their own docs, re-verified 2026-09-18.

INPUT LANGUAGES

18

Arabic among them

OUTPUT VOICES

16

11 English, 5 European

ARABIC VOICES

0

listed as coming soon

So a managed Arabic agent hears Arabic and answers in a European voice, and a judge who tests it hears exactly that.

CONSEQUENCE

So this is not the managed agent.

Transcription only from AssemblyAI, on the one model that takes Arabic. Everything after the transcript is ours, and every hop below was run against the live API.



WHAT THIS BUYS

One socket carries a whole conversation: two utterances, turn_order 0 then 1, zero reconnects. The free tier caps new streaming connections at 5 per minute, so a socket-per-turn design rate-limits itself after five exchanges.

WHAT IT COSTS

A browser cannot set an Authorization header on a WebSocket, so the token mint is public and cannot be made private. TTL and rate limits are speed bumps. The account spend cap is the only real bound.

FINDING 1

Numbers come back as words, not digits.

The audio said nine. The transcript says this.

WHAT THE SERVICE RETURNS

The image shows the Arabic word 'تسعة' (tase'ah) in a stylized, bold, light blue font. It is positioned on the left side of a comparison box.

one word, no digit anywhere in the turn

The image shows the digit '9' in a bold, light blue font. It is positioned on the right side of a comparison box, separated from the Arabic word by a vertical line.

what every downstream regex is looking for

```
re.search(r"\d{1,2}", transcript) -> None
```

Any agent matching `\d` finds nothing, books no time, and raises no error. It is a silent failure, not a crash.

FINDING 2

Partials are revised, not extended.

Captured live. Between partial 2 and partial 3 the service went back and changed a word it had already emitted. There is no retraction event.

PARTIAL 1

هل يمكنكم؟

PARTIAL 2

to the YEAR

هل يمكنكم تأجيل التصوير إلى السنة؟

PARTIAL 3

to NINE O'CLOCK

هل يمكنكم تأجيل التصوير إلى الساعة تسعة صباحاً؟

السنة became الساعة تسعة

Prefix-matched barge-in fires on words the caller never said, and the action cannot be taken back. Act on end_of_turn.

FINDING 3

Twilio's native 20 ms frame is rejected.

The socket closes. This refuted the design's central premise, which was to forward the phone bytes unchanged.

WHAT THE SOCKET SAID

```
error_code 3007: Input Duration Error: Input Duration Violation: 20.0 ms.  
                Expected between 50 and 1000 ms
```

```
socket closed with code 3007, deterministic, never retried
```

LEGAL FRAME DURATION



rejected

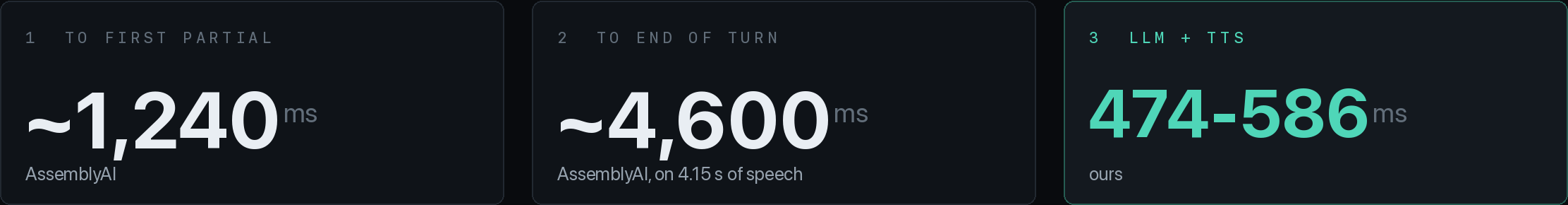
the legal window, every frame aggregated to 100 ms

Aggregation is mandatory and costs up to 100 ms. The bytes are still never resampled: mu-law in, mu-law out.

LATENCY

Three numbers, and blending them hides the one that moved.

Measured on the captured Arabic, a 4.15 s clip. Stage 2 is bounded below by how long the caller spoke, so it is compared to the speech duration, never to zero.



endpoint lag, caller stops to turn committed 575 – 775 ms

End of speech to a spoken Arabic answer is about 1.05 to 1.36 s. Not from a faster model: from not calling one.

WHAT THIS REPLACED

3,878 to 16,917 ms, measured over 7 real logged turns, essentially all of it the LLM. The agent is deterministic instead, so the rules path answers in 0.018 ms median.

THE VOICE IS FLAT WITH LENGTH

546 ms of speech costs 421 ms to render. 14,928 ms of speech costs 483 ms. The cost is process startup, not synthesis. Stage 3 marks COMPLETE audio, not a first byte.

THE DEMO

Every number on the panel is measured on that turn.

Arabic realtime voice agent

AssemblyAI v3 streaming socket · universal-3-5-pro · 16 kHz PCM s16le, 100 ms frames

● LISTENING

MODE **MOCK (REPLAY)**

CAPTURE -

VOICE AR-001 · MAJED

mock script finished. Press Replay to run it again.

MOCK

No server, no API key, no microphone. **Turn 0 is the real captured session** in fixtures/v3_session_arabic.jsonl, revision included, and its first two stage timings are the measured live figures from NOTES.md §3. Every agent reply and slot payload below was **captured from this repo's own /api/agent**. Turn 1 is a scripted continuation, and the invented timings are the only numbers here that are not measured, so they are marked **sim**.

Replay demo

Stop

Reset turn

no socket · no key · no microphone

TURN BUDGET

three stages, measured separately, never blended into one number

TIME TO FIRST PARTIAL

1102ms

from start of turn **SIM**

TIME TO END OF TURN

2722ms

from start of turn **SIM**

TIME TO REPLY

2880ms

after end of turn **SIM**

TURN	FIRST PARTIAL	END OF TURN	REPLY
1	1102 ms	2722 ms	2880 ms
0	1264 ms	4433 ms	4117 ms

CONVERSATION

logical Arabic, shaped by the browser. Nothing is pre-shaped or reversed in JS.

AGENT

أهلاً وسهلاً، معك استوديو بيكسلوجيك للتصوير. أي نوع تصوير تحتاج؟
فوتوغرافي، فيديو، مقابلة، أو تصوير منتجات؟

CALLER · TURN 0

هل يمكنكم تأجيل التصوير إلى الساعة تسعة صباحاً؟

AGENT

طيب. أي نوع تصوير تحتاج؟ فوتوغرافي، فيديو، مقابلة، أو تصوير منتجات؟

PARTIAL REVISION

partials are revised, not extended. A word already shown can be withdrawn.

1 REVISION THIS SESSION

PARTIAL · TURN 1

تصوير

PARTIAL · TURN 1

تصوير فيديو من فضلك

FINAL · TURN 1

END OF TURN

تصوير فيديو من فضلك

PARTIAL · TURN 0

هل يمكنكم؟

PARTIAL · TURN 0

WHAT IS ON SCREEN

- The partial stream, revisions marked
so the judge sees why acting on partials is unsafe rather than being told
- Raw Arabic beside the parsed slots
the number word becoming a value is on screen, not described
- A latency strip of three numbers
first partial, end of turn, reply, never blended into one
- Anything not measured on that turn is tagged sim
and the replay is labelled a replay

Fallback for a judge with no microphone and no Arabic speaker to hand:
?mock=1 replays the real captured session end to end, clearly labelled as a replay.

Syamjith NK

07

WHO THIS IS FOR

The failure is silent, so nobody files a bug.

A booking agent matching \d against Arabic finds nothing, books no time, and raises no error. It looks like a quiet day.

ASSEMBLYAI'S OWN ACCURACY TIERS

10-25%

the WER tier Arabic sits in

English sits under 10%

END TO END THROUGH THE LIVE API

12 of 12

values recovered by the parser

7 of 12 transcripts came back verbatim

THE OTHER FIVE WERE ORTHOGRAPHIC VARIATION, AND THE VALUE STILL CAME OUT RIGHT

ASKED	إلا ربعاً	والربع	خمسمئة	وأربعين	الاستوديو
HEARD	إلا ربع	وربع	خمسمائة	واربعين	الستوديو

A parser written against the one captured spelling would have looked perfect on the fixture and failed here. This is a synthetic speaker on a clean line: a floor on failure modes, not evidence of robustness on real callers.

LIMITS, STATED AS LIMITS

Every other entry will overclaim.

These are the gaps as they stand today, in the order they would bite someone who deployed this.

01 Accuracy on genuinely human Arabic is unmeasured.

The single biggest gap. Everything measured here has been synthetic or captured-synthetic audio. No dialect variety, no noise, no mobile codec.

02 Real microphone capture is unverified.

Chrome's fake device exercised the worklet, the resampler and the framing. It emits a tone, not speech. No Arabic has gone through a real microphone on the browser path.

03 The agent has no calendar.

It will happily confirm a slot that is already booked, and done writes nothing anywhere.

04 The voice is a system voice.

Intelligible, not warm. No measurement here defends it.

05 Mid-turn reconnect loses the first half of a sentence.

A new session has no memory of the old one. Untested live.

This slide is not modesty and it is not risk management. It is the strongest differentiator available.

WHERE TO LOOK

An Arabic voice agent is not an English one with the language code changed.

Everything on these slides was measured against the live API and is encoded as a test, so it cannot silently stop being true.

REPOSITORY

github.com/Syamjith-NK/arabic-voice-agent

MIT licensed. numpy and websockets; everything else is stdlib.

HOSTED DEMO

[URL NOT YET DECIDED]

hosting is an open decision in SUBMISSION.md section 9

BUILT DURING THE WINDOW

- v3 streaming client, measured against the live API
- deterministic booking agent with stateless resume
- web demo, driven in a real browser and screenshotted
- Arabic number-word parser
- Arabic speech with no vendor, no key and no bill
- the serverless shape: mint a token, take one turn

CHECKS THAT FAIL THE BUILD

487 agent **232** parser **97** stream **5** serverless

AssemblyAI Voice Agent Hackathon, lablab.ai